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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – FEB / MARCH 2025

Program	:	**B.E. – Information Science and Engineering**	**Semester**	:	**III**
Course Name	:	**Object Oriented Programming with Java**	**Max. Marks**	:	**100**
Course Code	:	**IS36 (Prior to 2021 Batch)**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Define classes and objects. Write their syntax and examples. Develop a java program for a given scenario. A triangle class with the following data members breadth and length of type double. Perform constructor overloading to initialize data members and calculate the area of a circle using findArea() method. Display an appropriate message.	CO1	(10)
	b)	Define constructor. Explain in detail Default Constructor and Parameterized Constructor with suitable Java program.	CO1	(10)
2.	a)	Illustrate with the help of an example the use of for loop and for each loop in Java. Write a java program to search the given key.	CO1	(10)
	b)	What are classes and objects? Define a class Triangle with an appropriate methods and data members in this class to compute the area of a rectangle. Write main method to test the Triangle class.	CO1	(10)

UNIT - II

3.	a)	With an example of Java program write the significance of static variable, static method and static block.	CO2	(06)
	b)	Compare and contrast method overloading and method overriding with suitable examples?	CO2	(08)
	c)	Write Java program to find the factorial of a number using recursion. Take the number from command line.	CO2	(06)
4.	a)	Describe access control. Show a code snippet to access the members of class derived by default access, public, private and protected.	CO2	(08)
	b)	Is overloading of constructor is allowed in Java? Justify your answer with the help of a suitable program.	CO2	(06)
	c)	With the Java program, explain the two uses of 'super' with respect to inheritance.	CO2	(06)

UNIT - III

5.	a)	What is Run Time Polymorphism? With the help of an example explain dynamic method dispatch.	CO3	(06)
	b)	Illustrate the use of interfaces in Java with suitable Java program.	CO3	(08)
	c)	What are packages? How to create them? Explore How to access packages from another package?	CO3	(06)
6.	a)	Discuss usage of final keyword in inheritance with an example.	CO3	(06)
	b)	Justify whether Java supports Multiple inheritance through classes and interfaces.	CO3	(08)
	c)	Distinguish between abstract class and interfaces.	CO3	(06)

UNIT- IV

7.	a)	Distinguish between throw and throws keyword with the help of an example.	CO4	(10)
	b)	Explore all the possible cases of Java finally block while handling exceptions with suitable example.	CO4	(10)
8.	a)	What are exceptions? How to handled them in Java? Illustrate with an example java program.	CO4	(08)
	b)	Describe chained exception with an example.	CO4	(06)
	c)	Demonstrate multiple catch clause with syntax and example program.	CO4	(06)

UNIT - V

9.	a)	List and Explain the methods of List Interface.	CO5	(10)
	b)	Define array list. With the help of program explain how addition and deletion of elements works in array list.	CO5	(10)
10.	a)	Define autoboxing and unboxing with respect to Integer and character class with an example.	CO5	(10)
	b)	Write a JAVA program using LinkedList class to perform the following operations - to create an empty list, add {100,200,300} these elements to list, add element 5 to start of a list, add 400 to end of a list, display only first element, remove first element and last elements of a list, display elements of an array after every operation.	CO5	(10)

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